

# SIH26047 — Real Slide Content Draft

Content for the actual 6-slide official submission structure, not a generic pitch-deck shape. Structure verified via web search against the real AICTE/SIH format on 13 Sep 2026 (sources on the last page) — a strict 6-slide PDF, fixed template, one topic per slide, points not paragraphs. Every fact below was checked against the live code or the PS's own text this session. This is content to paste into the official template, not a replacement for it.

## SLIDE 1

### Title Slide

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|--------------------------------|---------------------------------|
| <b>Team Name</b>               | Inayat                          |
| <b>Problem Statement ID</b>    | SIH26047                        |
| <b>Problem Statement Title</b> | Patient Case-Taking Software    |
| <b>Theme</b>                   | MedTech / BioTech / HealthTech  |
| <b>Organization</b>            | Ministry of Ayush               |
| <b>Department</b>              | All India Institute of Ayurveda |
| <b>Category</b>                | Software                        |

Add team members' names/roles and college name/logo per your own institute's requirement — not tracked in this repository.

## Problem Statement / Idea Overview

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### **Problem, restated (in the team's own words)**

Indian OPD consultations average roughly two minutes — among the shortest in the world — yet a good clinical history alone accounts for 70–80% of correct diagnoses, before any exam or test. Patients arrive with fragmented, unstructured paper records from prior visits that the physician must manually sort through inside that same compressed window. AYUSH institutions carry an even heavier history-taking load: proper Ayurvedic assessment (Dashavidha Pariksha) is a far more extensive framework than allopathic intake, and there is no way to capture it inside a 2-minute slot.

### **Who this actually affects**

OPD patients at high-volume government and AYUSH hospitals — specifically the elderly, rural, low-literacy, and first-visit patients existing mobile-health apps already exclude by requiring smartphone fluency and pre-enrolment. And the treating physicians themselves, who lose consultation time to repeated basic questioning instead of examination and reasoning.

### **Why this hasn't been solved already**

Hospital registration systems capture only demographics and a token number, never clinical history. Tele-triage apps assume a literate, connected, pre-enrolled user. Manual nurse-led triage desks don't scale to thousands of daily patients and just move the same bottleneck earlier in the queue.

### **Scale**

Tertiary government hospitals routinely register 4,000–10,000 OPD patients per day.

## **Proposed Solution / Approach**

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### **Elevator pitch**

Inayat lets a patient describe what's wrong — by typing, speaking in their own language, or photographing a prior prescription — and turns that into a structured, physician-ready history summary before they ever enter the consultation room.

### **Key features**

- A real adaptive interview, not a static form: an eight-question SOCRATES framework for pain complaints, with separate real question sets for respiratory/GI/dermatological/general complaints, branching by what the patient actually reports.
- A deterministic safety net beneath the AI layer: red-flag detection and guideline verification that can only escalate a case's priority, never quietly lower one — and every part of the pipeline (speech recognition, speech synthesis, triage reasoning, history drafting) has a zero-network fallback, so the system degrades honestly instead of going down.
- Multi-document OCR with chronological ordering, so multiple prior prescriptions and reports reach the physician as one organized timeline, not a pile of separate images.
- Real trilingual support — English, Hindi, Telugu — for voice input, voice output, and the case summary itself, not just interface labels.
- An AYUSH history mode that only asks the patient what a patient can actually self-report (constitution, lifestyle, tolerance) and leaves physical-examination parameters for the physician, rather than asking a kiosk to fake a clinical exam.

### **How this differs from an existing solution or a typical AI wrapper**

Most conversational-intake prototypes are a thin interface over a single API call: if that call fails, the product fails. Inayat's safety-relevant logic (red-flag detection, guideline escalation) is deterministic and works with zero network access; the LLM layer is a real enhancement when available, not the only thing holding the system up.

## Technical Feasibility & Tech Stack

|                               |                                                                                                                                                                                                                                       |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Frontend</b>               | HTML / CSS / vanilla JavaScript — no framework dependency, light enough for low-end devices and slow connections.                                                                                                                     |
| <b>Backend</b>                | Python, FastAPI.                                                                                                                                                                                                                      |
| <b>Database</b>               | SQLite (persisted case records).                                                                                                                                                                                                      |
| <b>AI / ML</b>                | Anthropic Claude for triage reasoning and history drafting when configured, with a real deterministic fallback when it isn't; scikit-learn TF-IDF similarity for guideline-based escalation; Tesseract OCR for document digitization. |
| <b>Speech</b>                 | Bhashini (Government of India / MeitY) as the preferred speech engine; PocketSphinx (offline ASR, English) and espeak-ng (offline TTS, English/Hindi/Telugu) as genuine zero-network fallbacks.                                       |
| <b>Government integration</b> | ABDM (Ayushman Bharat Digital Mission) sandbox — OTP-based enrollment with real RSA-OAEP encryption matching ABDM's published spec.                                                                                                   |

### Feasibility statement

This is not a mockup: every component above is real, running code, deployed and reachable at a live URL, with the offline/zero-credential paths independently verified end to end. The 36-hour build window this table describes is scoped to hardening and extending an existing, working system — not building it from nothing.

Add your live deployment URL here at submission time.

## Impact & Benefits

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Framed as Low / Medium / High, not invented figures — the official template's own guidance is to avoid fabricated market-size or adoption numbers.

|                             |               |                                                                                                                                                                                                                     |
|-----------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Social impact</b>        | <b>High</b>   | Directly serves the PS's own named excluded populations — elderly, rural, low-literacy, first-visit patients — via voice input in their own language and a working system even with no internet or paid API access. |
| <b>Economic impact</b>      | <b>Medium</b> | Less physician time spent re-asking basic history questions frees consultation time for actual examination and reasoning, inside a system that scales without per-patient API cost.                                 |
| <b>Environmental impact</b> | <b>Low</b>    | Secondary benefit only: digitizing and timelining paper prescriptions/reports reduces repeated physical document handling; not a primary design goal.                                                               |
| <b>Scalability</b>          | <b>High</b>   | The zero-API deterministic fallback means the system keeps functioning at high patient volumes even without paid infrastructure scaling in lockstep — a real constraint at 4,000+ patients/day.                     |

## Research & References / Team

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### Key sources

- Classical clinical-history teaching: a good history alone yields 70–80% of correct diagnoses, before exam or test.
- BMJ Open, 2017 — 67-country study placing India's average primary-care consultation at just over 2 minutes, among the shortest studied.
- Dashavidha Pariksha literature, cross-checked across independent sources: a published Zenodo review, AyurVAID's clinical diagnostics reference, the Journal of Ayurveda and Integrated Medical Sciences, and CCRAS — the Ministry of Ayush's own clinical research body.
- ABDM's own published API and RSA-OAEP encryption specification, implemented and tested directly rather than assumed from documentation alone.

### Existing solutions studied, and how this differs

- Hospital registration kiosks — demographics and a token number only, no clinical history.
- Mobile tele-triage apps — require smartphone literacy and pre-enrolment, excluding the population this PS is actually about.
- Manual nurse-led triage desks — human-resource-limited, doesn't scale, moves the same bottleneck earlier rather than removing it.

### Team

Inayat — Urdu for grace: attentive care shown by one with more knowledge or standing toward one in need.

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Slide structure sourced from a community transcription of the SIH 2026 PPT submission template ([github.com/tharsan1305/SIH-2026](https://github.com/tharsan1305/SIH-2026)) and cross-checked against multiple independent search results describing the same AICTE 6-slide format, 13 Sep 2026 — not the primary government portal directly, which was unreachable from this environment. Confirm against your institute's SIH SPOC or the official portal before final submission, the same standing caveat this project's own PS documentation already carries.